Cervical Cancer Detection

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Introduction

The “Brown Multicellular ThinPrep” (BMT) dataset contains 600 multicellular liquid Pap smear images that can be analyzed to assist in the detection of cervical cancer. These cytology images are categorized into three main classes, with 200 images each: Negative for Intraepithelial Lesion or Malignancy (NILM), Low-grade Squamous Intraepithelial Lesion (LSIL), and High-grade Squamous Intraepithelial Lesion (HSIL). These categories represent normal findings (NILM), mild abnormalities (LSIL), and severe abnormalities (HSIL). Accurate classification of these images can ensure early detection of cervical cancer.

The goal of this project is to develop and evaluate deep learning models for classifying Pap smear images into NILM, LSIL, and HSIL categories using the BMT dataset, and to compare performance with the results reported in the reference paper. The models EfficientNet-B4, and ConvNeXt-Small, are explored to assess their effectiveness on this task. Model performance is evaluated using precision, recall, F1-score, accuracy, AUC, and confusion matrices.

Materials and Methods

Dataset

The BMT dataset consists of 600 multicellular ThinPrep liquid Pap smear images, evenly distributed across three classes: NILM (200), LSIL (200), and HSIL (200). Images were stored in a single flat directory with class labels embedded in filenames (e.g., "HSIL (1).jpg", "NIL (43).jpg").

The label "NIL" was mapped to "NILM" for consistency with the used terminology. Following the split used in the reference paper, images were partitioned using a stratified 60/20/20 train/validation/test split. With 200 images per class, this yields 120 training, 40 validation, and 40 test images per class — totaling 360 training, 120 validation, and 120 test images. A fixed random seed of 42 was used to ensure reproducibility. All images were resized to a fixed input resolution and normalized to match the requirements of the pretrained models.

Data Set Up

For the EfficientNet-B4 model, a 70/15/15 train/validation/test split was initially used. This split previously performed well during a local CPU-based experiment using VGG19, where an accuracy of 71.11% was achieved, exceeding the 64.16% reported in the reference paper.

However, EfficientNet-B4 produced unsatisfactory results under this setup. When transitioning to ConvNeXt-Small, the data split was adjusted to 60/20/20 to match the methodology used in the reference paper and allow for more direct comparison.

Training Procedures

Both models were trained using a two-stage fine-tuning strategy. During the first 8 epochs, only the classifier head was trained at a learning rate of 3×10^{-4}. In Stage 2 (epochs 9 onward),

the backbone was unfrozen and trained using discriminative learning rates: backbone at 1×10^{-5} and classifier head at 5×10^{-5}. A 3-epoch linear warmup was applied at the start of Stage 2, followed by cosine annealing for the remaining epochs.

The AdamW optimizer was used with weight decay of 1×10^{-4}. The loss function was cross-entropy with label smoothing of 0.1 and class weights inversely proportional to class frequency. Mixup augmentation (α=0.3) was applied during training. Gradient clipping at 0.5 was used to prevent instability during fine-tuning. A weighted random sampler was used during training to address class imbalance. Models were trained for 70 epochs (ConvNeXt-Small) and 50 epochs (EfficientNet-B4), with checkpoints saved based on best validation macro F1-score.

Evaluation

Models were evaluated on the held-out test set using the following metrics: accuracy, macro-averaged F1-score, per-class F1-score (NILM, LSIL, HSIL), macro-averaged AUC-ROC (one-vs-rest), and confusion matrices. Test-time augmentation (TTA) with 5 augmented passes was also applied, with final predictions obtained by averaging softmax probabilities across all passes. This basically applies random flips and crops to predictions. Initially a local environment was considered, but the final two models were trained on Google Colab using an NVIDIA T4 GPU.

Results

Training Behavior

Both models followed a two-stage training trajectory consistent with the fine-tuning strategy. During Stage 1 (epochs 1–8), the frozen backbone allowed rapid improvement of the classifier head, with validation loss decreasing steadily. Upon backbone unfreezing at epoch 8, early experiments with ConvNeXt-Small produced NaN losses due to learning rate instability. This also happened with EfficientNet-B4 a little after unfreezing, around epoch 18 specifically. This was resolved by introducing discriminative learning rates and a linear warmup, after which training proceeded stably for all 70 epochs. In both models, the best validation F1 checkpoint was restored for test evaluation.

Performance

Table 1 summarizes test set performance for both models compared to the paper baselines. Per-class F1 and approximate AUC values for the paper baselines are read directly from Fig. 5 of the reference paper; overall accuracy is as reported in the paper text.

Model	Accuracy	Macro F1	AUC-ROC	F1 NILM	F1 LSIL	F1 HSIL
VGG19 (paper)	0.6416	0.6333	≈0.81	0.65	0.52	0.73
ResNet50 (paper)	0.7416	0.7367	≈0.91	0.71	0.68	0.82
VGG19 (local CPU)	0.7111	0.7106	0.8904	0.7368	0.6786	0.7164
EfficientNet-B4	0.6778	0.6619	0.8415	0.7302	0.5333	0.7222
EfficientNet-B4 + TTA	0.6889	0.6727	0.8589	0.7742	0.5217	0.7222
ConvNeXt-Small	**0.8833**	**0.8851**	**0.9830**	**0.9333**	**0.8750**	**0.8471**
ConvNeXt-Small + TTA	0.8667	0.8675	0.9809	0.8947	0.8608	0.8471

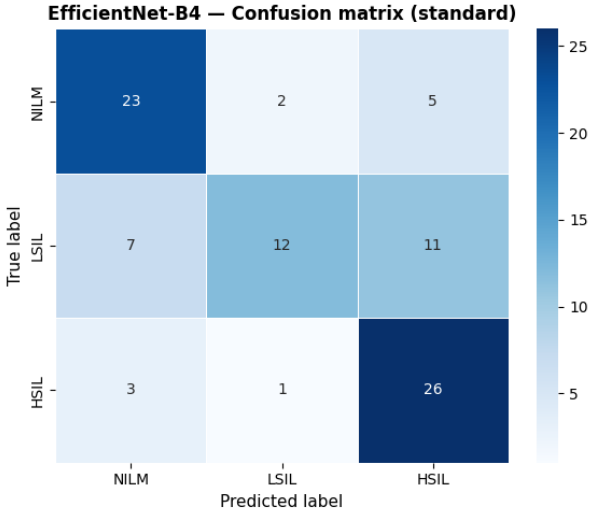
Table 1. Test set performance. Paper baseline AUC and per-class F1 values are estimated from Fig. 5 of the reference paper. EfficientNet-B4 used a 70/15/15 split (90 test images); ConvNeXt-Small used the 60/20/20 split (120 test images) matching the paper.

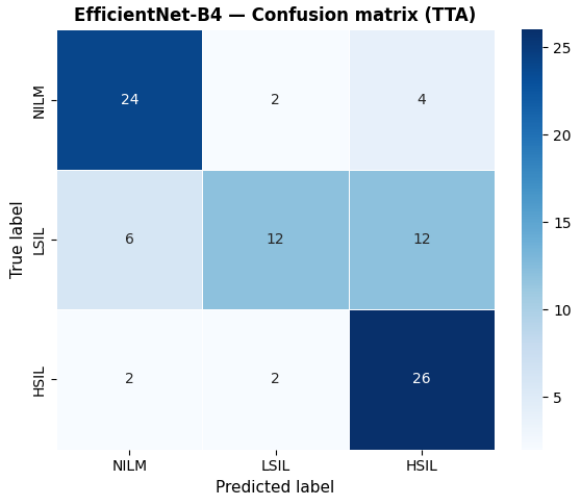
ConvNeXt-Small achieved 88.33% accuracy and a macro F1-score of 0.8851, representing a 14.17 percentage point improvement over the best paper baseline (ResNet50: 74.16%). The macro-AUC-ROC of 0.9830 substantially exceeds the approximate 0.91 readable from the ResNet50 ROC curves in the reference paper. EfficientNet-B4 achieved 67.78% accuracy (standard) and 68.89% with TTA on the 70/15/15 split, which are broadly consistent with the ResNet50 baseline given the smaller and differently split test set. The substantial gap between the two models is primarily attributable to the training instability fix (discriminative learning rates) applied only to ConvNeXt-Small, the larger test set providing more stable metrics, and the inherent architectural advantages of ConvNeXt-Small.

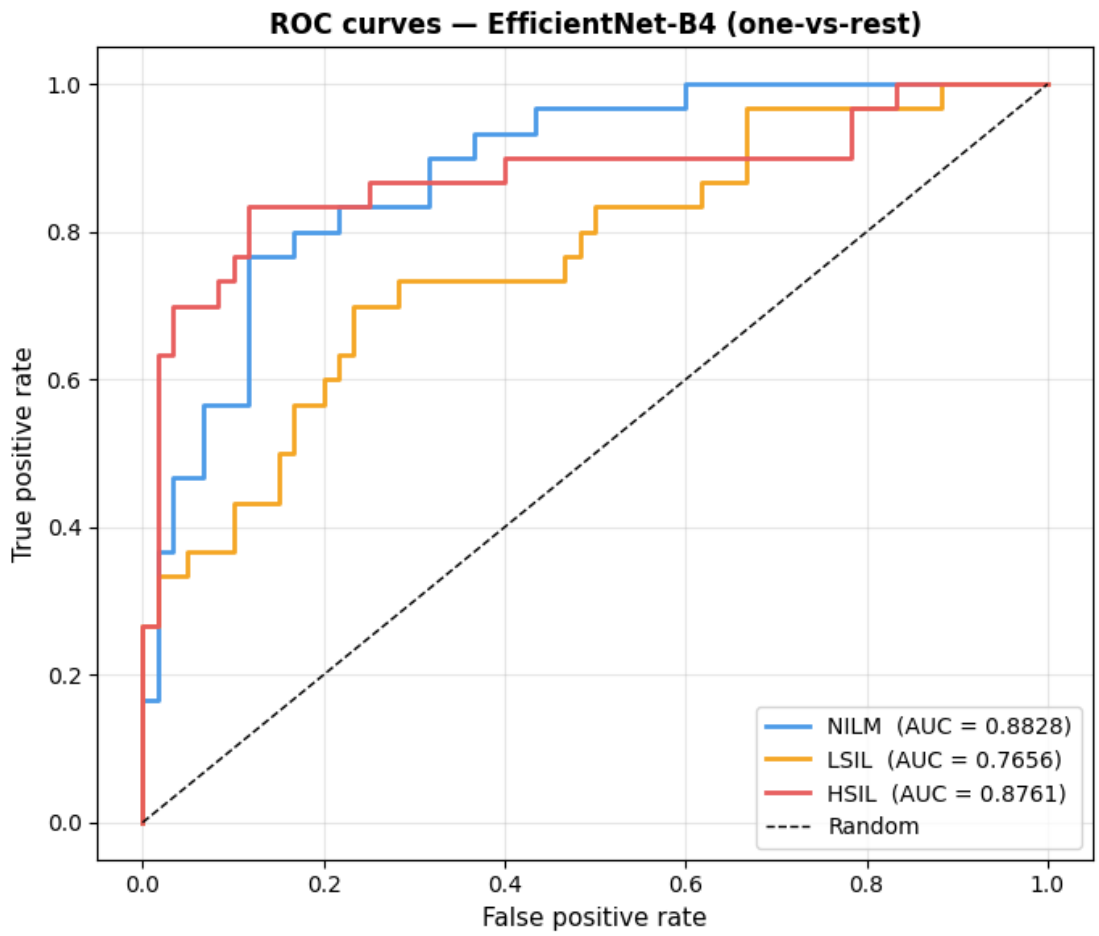
LSIL was the most challenging class for all models. ConvNeXt-Small achieved F1 0.8750 for LSIL, compared to 0.68 for ResNet50 and 0.52 for VGG19 in the reference paper. NILM achieved the highest F1 (0.9333) with 100% precision for ConvNeXt-Small, indicating no false positives

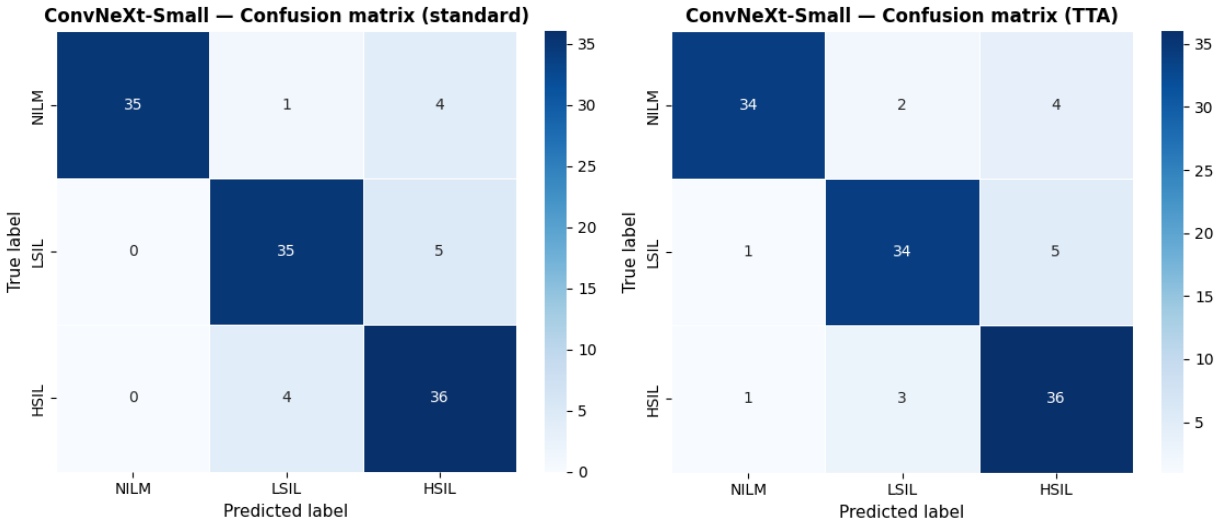
for the normal class — a clinically valuable property. Test-time augmentation did not improve performance for ConvNeXt-Small, with accuracy decreasing from 88.33% to 86.67%, while TTA provided a marginal 1.11% accuracy improvement for EfficientNet-B4.

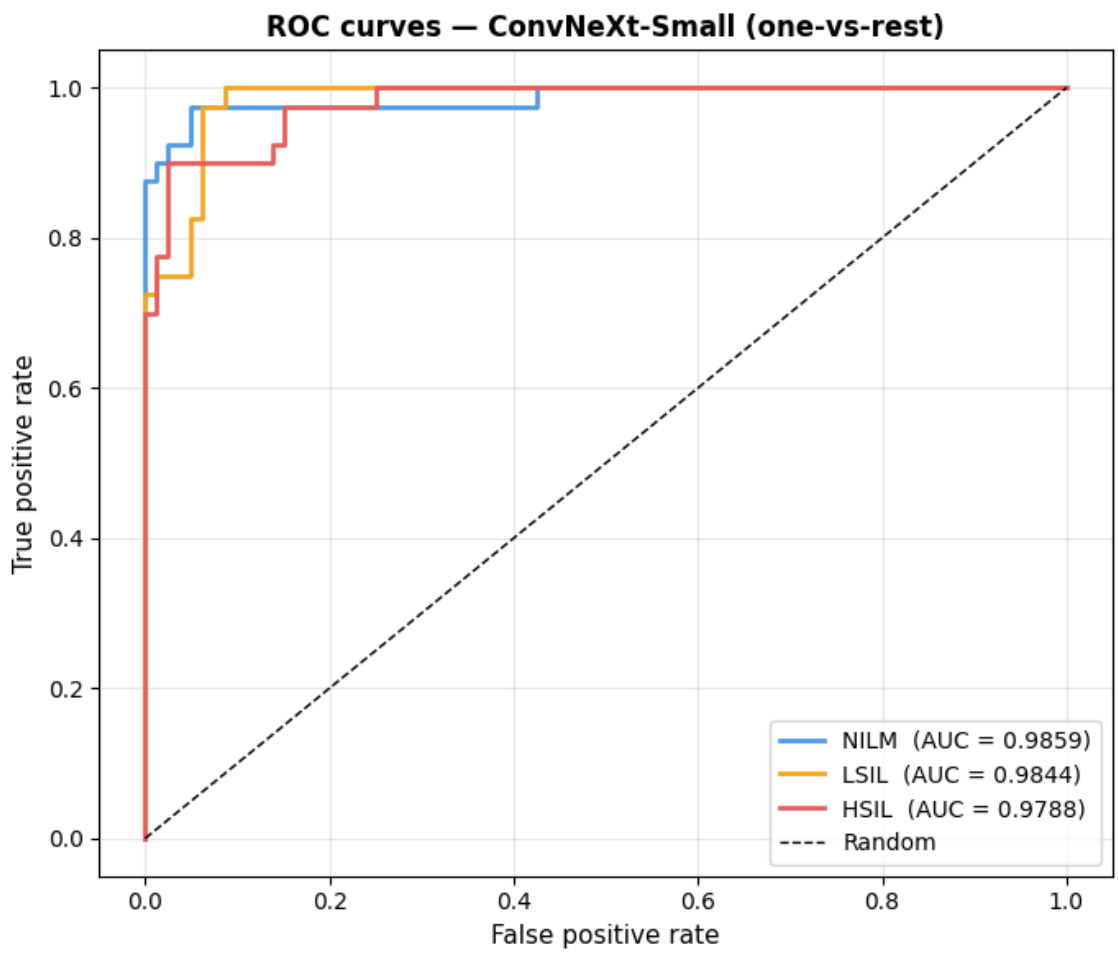
Confusion Matrices and ROC Curves








ConvNeXt-Small — Confusion matrix (standard)
True label
Predicted label
NILM
LSIL
HSIL
35
1
4
0
35
5
0
4
36
ConvNeXt-Small — Confusion matrix (TTA)
34
2
4
1
34
5
1
3
36


ROC curves — ConvNeXt-Small (one-vs-rest)
True positive rate
False positive rate
NILM (AUC = 0.9859)
LSIL (AUC = 0.9844)
HSIL (AUC = 0.9788)
Random

Discussion

This study evaluated EfficientNet-B4 and ConvNeXt-Small for Pap smear cytology classification on the BMT dataset. ConvNeXt-Small achieved 88.33% accuracy and a macro F1-score of 0.8851, substantially exceeding the best published baseline on this dataset (ResNet50: 74.16%) by 14.17 percentage points. EfficientNet-B4 performed more modestly at 67.78% on the 70/15/15 split, comparable to the ResNet50 baseline when accounting for differences in data splits. This lower performance is primarily attributed to training instability that was not fully resolved at the time of evaluation.

The performance gap between EfficientNet-B4 and ConvNeXt-Small in this study highlights the critical importance of fine-tuning strategy over raw architectural capability. EfficientNet-B4 is a stronger architecture than ResNet50 by most benchmarks, yet it did not improve on it here due to gradient instability at unfreezing. ConvNeXt-Small, trained with discriminative learning rates and a linear warmup, was able to stably adapt its pretrained representations to the cytology domain. This underscores that for small medical imaging datasets, training procedure is as important as model selection.

The LSIL class remained the most challenging across all models. The reference paper's ResNet50 also showed lowest F1 for LSIL (0.68), and this study's ConvNeXt-Small improved that to 0.8750, a gain of 0.1950 F1 points. The elastic deformation and random erasing augmentations added in this work were specifically intended to improve LSIL generalization by simulating natural variation in liquid Pap preparation.

TTA did not improve results for ConvNeXt-Small, with accuracy decreasing marginally from 88.33% to 86.67%. This is explained by the model's already-high confidence: averaging over

augmented views introduces noise where the standard model was correct with certainty. The NILM class illustrates this clearly, where standard inference achieves 100% precision, which TTA degrades to 94.4% by occasionally cropping discriminative regions. TTA is most beneficial for uncertain models and since ConvNeXt-Small had high performance, the performance metrics decreased with TTA.

A key limitation of this study is the relatively small dataset size (600 images), which may limit generalization to real-world clinical settings.

Conclusion

This project explored the use of deep learning models for classifying Pap smear cytology images into NILM, LSIL, and HSIL categories using the BMT dataset. Among the evaluated models, ConvNeXt-Small achieved the best performance, reaching 88.33% accuracy and a macro F1-score of 0.8851, significantly outperforming the VGG19 and ResNet50 baselines reported in the reference paper. EfficientNet-B4 achieved more modest results, largely due to training instability during fine-tuning, highlighting the importance of training strategy when working with small medical imaging datasets.

Overall, the results demonstrate that modern architectures such as ConvNeXt can provide substantial improvements over traditional CNN models when properly fine-tuned. The study also showed that techniques such as discriminative learning rates and learning rate warmup are critical for stable training. While LSIL remained the most challenging class, performance improvements across all classes indicate that deep learning models can effectively support automated cytology classification.

Future work could focus on further hyperparameter tuning, applying the same stabilized training approach to other architectures, and exploring larger datasets or ensemble methods to improve generalization. These improvements could help move toward more reliable and clinically applicable AI-assisted diagnostic systems.

References

Welch, E.C., Lu, C., Sung, C.J. et al. BMT: A Cross-Validated ThinPrep Pap Cervical Cytology Dataset for Machine Learning Model Training and Validation. Sci Data 11, 1444 (2024). https://doi.org/10.1038/s41597-024-04328-3